

CS471 – Lesson 21 Solutions Key

Probabilistic Systems: Bayesian Inference

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Air-Strike Mission Bayesian Network Formulation

Scenario: Network $W \rightarrow V$, $W \rightarrow C$, $V \rightarrow M$, $C \rightarrow M$. Query is $\mathbf{P}(M \mid W = \text{Severe})$.

a. Variable Partitioning:

Solution

- Query Variable (Q): **M** (Mission Outcome)
- Evidence Variable(s) (E): **W** = Severe
- Hidden Variable(s) (H): **V** (Visibility) and **C** (SATCOM)

b. Summation Formula:

Solution

$$P(M = \text{Success} \mid W = \text{Severe}) = \sum_{v \in \{G, P\}} \sum_{c \in \{R, D\}} P(v \mid \text{Sev}) \cdot P(c \mid \text{Sev}) \cdot P(\text{Success} \mid v, c)$$

(Note: Because W is conditioned on, V and C factor independently).

c. Numerical Calculation:

Solution

Evaluate all 4 combinations of (V, C) given $W = \text{Severe}$:

- $V = \text{Good}, C = \text{Reliable}$: $0.30 \times 0.40 \times 0.95 = 0.1140$
- $V = \text{Good}, C = \text{Disrupted}$: $0.30 \times 0.60 \times 0.60 = 0.1080$
- $V = \text{Poor}, C = \text{Reliable}$: $0.70 \times 0.40 \times 0.50 = 0.1400$
- $V = \text{Poor}, C = \text{Disrupted}$: $0.70 \times 0.60 \times 0.10 = 0.0420$

$$P(M = \text{Success} \mid W = \text{Severe}) = 0.1140 + 0.1080 + 0.1400 + 0.0420 = \mathbf{0.4040} \quad (40.40\%)$$

2. Normalization and Diagnostic Reasoning

Unnormalized weights: $P(\text{Clear}, \text{Ev}) = 0.0646$, $P(\text{Severe}, \text{Ev}) = 0.0292$.

a. Normalization Constant α :

Solution

$$P(\text{Evidence}) = 0.0646 + 0.0292 = 0.0938$$

$$\alpha = \frac{1}{0.0938} \approx \mathbf{10.6610}$$

b. Normalized Posterior Distribution:

Solution

$$P(W = \text{Clear} \mid \text{Evidence}) = \frac{0.0646}{0.0938} \approx \mathbf{0.6887} \quad (68.87\%)$$

$$P(W = \text{Severe} \mid \text{Evidence}) = \frac{0.0292}{0.0938} \approx \mathbf{0.3113} \quad (31.13\%)$$

$$\text{Sum} = 0.6887 + 0.3113 = 1.0000.$$

Part 2: Student Independent Practice Solutions

Autonomous UAV Defense Network

Structure: $J \rightarrow S \rightarrow L$.

1. Causal Forward Inference $P(L = 1 \mid J = 1)$:

Solution

Marginalize out $S \in \{0, 1\}$:

$$P(L = 1 \mid J = 1) = \sum_{s \in \{0,1\}} P(S = s \mid J = 1)P(L = 1 \mid S = s)$$

- $S = 1$: $P(S = 1 \mid J = 1)P(L = 1 \mid S = 1) = 0.80 \times 0.90 = 0.7200$
- $S = 0$: $P(S = 0 \mid J = 1)P(L = 1 \mid S = 0) = (1 - 0.80) \times 0.05 = 0.20 \times 0.05 = 0.0100$

$$P(L = 1 \mid J = 1) = 0.7200 + 0.0100 = \mathbf{0.7300} \quad (73.00\%)$$

2. Diagnostic Root-Cause Inference Unnormalized Weights:

Solution

a. For $J = 1$:

$$P(J = 1, L = 1) = \sum_{s \in \{0,1\}} P(J = 1)P(S = s \mid J = 1)P(L = 1 \mid S = s)$$

$$P(J = 1, L = 1) = 0.10 \times [0.80(0.90) + 0.20(0.05)] = 0.10 \times [0.72 + 0.01] = \mathbf{0.0730}$$

b. For $J = 0$:

$$P(J = 0, L = 1) = \sum_{s \in \{0,1\}} P(J = 0)P(S = s \mid J = 0)P(L = 1 \mid S = s)$$

$$P(J = 0, L = 1) = 0.90 \times [0.10(0.90) + 0.90(0.05)] = 0.90 \times [0.090 + 0.045] = 0.90 \times 0.135 = \mathbf{0.1215}$$

3. Posterior Normalization:

Solution

Total evidence probability $P(L = 1) = 0.0730 + 0.1215 = 0.1945$.

$$\alpha = \frac{1}{0.1945} \approx \mathbf{5.1414}$$

$$P(J = 1 \mid L = 1) = \frac{0.0730}{0.1945} \approx \mathbf{0.3753} \quad (37.53\%)$$

$$P(J = 0 \mid L = 1) = \frac{0.1215}{0.1945} \approx \mathbf{0.6247} \quad (62.47\%)$$

4. Tactical Interpretation:

Solution

$P(J = 1 \mid L = 1) = 37.53\%$ is much lower than 90% due to the ****Base Rate Effect****: Cyber Jamming is rare ($P(J = 1) = 10\%$). Even though jamming severely degrades tracking, unjammed missions occur 90% of the time, and their baseline sensor noise (13.5% combined loss rate) produces a larger raw volume of target losses (0.1215) than jamming does (0.0730).